

Question Paper

Exam Date & Time: 19-Jun-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FOURTH SEMESTER B.TECH. (INFORMATION TECHNOLOGY) DEGREE EXAMINATIONS - JUNE 2024
SUBJECT: MAT 2226/MAT_2226 - ENGINEERING MATHEMATICS - IV

Marks: 50

Duration: 180 mins.

Answer all the questions.

Missing data may be suitably assumed.

1A) Six people toss a fair coin one after the other until a player obtains a head. The game is won by the player who throws head first. Find the probability of success of the 4th player. (4)

1B) Two defective tubes get mixed up with 6 good ones. The tubes are tested one by one, until both defective tubes are found. What is the probability that the last defective tube is obtained on (a) the 2nd test, (b) the 3rd test. (3)

1C) Let X be a discrete random variable having the probability mass function given by: (3)

x	1	2	3	4	5
$P(X = x)$	1/15	2/15	3/15	4/15	5/15

(i) Check whether the given function is a valid probability mass function.

(ii) Calculate $P(1/2 < X < 5/2 \mid X > 1)$.

2A) Find the mean and variance of a random variable X which is uniformly distributed in the interval $(-a, a)$. (4)

2B) An aircraft knows that 5% of the people making reservation on a certain flight will not show up. Consequently, their policy is to sell 52 tickets for the flight that can only hold 50 passengers. What is the probability that there will be a seat available for every passenger who turns up? (3)

2C) Let X has binomial distribution with parameter $n=10$ and $p \in \{p : p = \frac{1}{4}, \frac{1}{2}\}$. (3)

$H_0: p = \frac{1}{2}$. Is rejected and $H_1: p = \frac{1}{4}$ is accepted. If the observed values of X_1 a random sample of size 1 is less than or equal to 3, find the power function of the test.

3A) Let (X, Y) have the joint pdf (4)

$$f(x, y) = \begin{cases} 2y, & 0 < x < 1, 0 < y < 1 \\ 0 & \text{otherwise.} \end{cases}$$

(i) Find the mean and variance of X and Y .

(ii) Find $E(X \mid Y)$.

3B) Urn A contains 5 black balls, 6 white balls. Urn B contains 8 black balls and 4 white balls. Two balls are transferred from B to A and then a ball is drawn from A. (3)

i) What is the probability that this ball is white?

ii) Given that the ball is drawn is white. what is the probability that one black ball was transferred to A?

3C) The survival time X (in terms of weeks) of a male mouse exposed to radiation has Gamma distribution with PDF (3)

$$f(x) = \begin{cases} \left(\frac{1}{15}\right)^3 \frac{x^2 e^{-\frac{x}{15}}}{\Gamma(3)}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

Find the probability that a mouse survives:

- i) Between 60 and 120 weeks
- ii) At least 30 weeks

4A) Suppose X and Y are independent random variables. X takes values 2, 5, 7 with probability $\frac{1}{2}, \frac{1}{4}, \frac{1}{4}$ respectively. Y takes the value 3, 5 with probability $\frac{1}{3}, \frac{2}{3}$. (4)

- i) Find joint probability distribution of X and Y .
- ii) Find probability distribution of $Z = X + Y$.

4B) The probability that a news reader commits no mistakes in reading the news during one broadcast is $\frac{1}{e^3}$. Find the probability that in ten news broadcasts he commits (3)

- i) Only one mistake
- ii) More than 3 mistakes
- iii) At most 2 mistakes.

4C) Assume that the average life span of computers produced by the company is 2040 hours with standard deviation 60 hours. Find the expected number of computers, whose life span is: (3)

- i) more than 2150 hours,
- ii) less than 1950 hours,
- iii) between 1920 and 2160 hours

from a pool of 2000 computers, assuming that life span, X , is normally distributed.

5A) According to the ideal grading curve, the percentages of students receiving grades A+, A, B, C, D, E, and F should be, respectively, 0.1, 2.2, 13.6, 34.1, 34.1, 13.6, and 2.3. In an exam with 2000 students, the numbers of students getting grades A+, A, B, C, D, E, and F are, respectively, 3, 48, 290, 642, 724, 280, 13. Test whether the grade distribution fits the ideal grading curve, using Chi-square test at 5% significance level. (4)

5B) Let $(X_1, X_2, \dots, X_n)$ be a random sample from normal distribution $N(0, \theta)$, where (3)

$0 < \theta < \infty$. Show that $Y = \frac{\sum_{i=1}^n X_i^2}{n}$ is an unbiased and consistent estimator for θ .

5C) The mean life length of a certain cutting tool is 41.5 hours with a standard deviation of 2.5 hours. (3)

What is the probability that a random sample of size 50 drawn from this population will have a sample mean between 40.5 and 42 hours.

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Exam Date & Time: 25-Jul-2022 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

ENGINEERING MATHEMATICS IV [MAT 2257]

Marks: 50

Duration: 180 mins.

Descriptive Questions

Answer all the questions.

Section Duration: 180 mins

- 1) Solve the following PDE $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = -81xy$, $0 < x < 1, 0 < y < 1$,
 A) $u(0, y) = u(x, 0) = 0$ and $u(1, y) = u(x, 1) = 100$ taking $h = \frac{1}{3}$. (3)
- B) Solve $y'' + xy = 1$, $y(0) = 0$, $y'(1) = 1$ with $h = 0.5$ by finite difference method. (3)
- C) Solve the wave equation $u_{tt} = u_{xx}$, $0 < x < 1, t > 0$, $u(x, 0) = \frac{\partial u}{\partial t}(x, 0) = 0$,
 $u(0, t) = 0$ and $u(1, t) = 100 \sin \pi t$. Compute u for 4-timesteps with $h = \frac{1}{4}$. (4)
- 2) Fit a curve $y = ax^2 + bx + c$ for the following data by the method of
- | | | | | |
|---|----|---|---|---|
| x | -1 | 0 | 1 | 2 |
| y | -2 | 1 | 2 | 4 |
- A) principle of least square. (3)
- B) Solve the difference equation using Z-transform method
 $y_{n+2} - 3y_{n+1} - 10y_n = 0, y_1 = 0, y_0 = 1$. (3)
- C) Solve the difference equation $y_{n+2} - 7y_{n+1} - 8y_n = 2^n [n]^2$. (4)
- 3) A survey shows 56% of all residents in a city have a workplace retirement plan, 68%
 Have health insurance, and 49% have both benefits. At random one worker is selected. (3)
- A) a. What is the probability that he has neither health insurance nor a retirement plan?

- b. What is the probability that he has health insurance if he has the retirement plan?
- c. Are having health insurance and a retirement plan independent events? Explain .

B) Suppose the pdf of a random variable X is given as

$$f(x) = \begin{cases} \frac{1-x}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

Find $E(X)$ and $V(X)$.

C) The contents of urns I, II, III are as follows:

1 white, 2 black and 3 red balls,

2 white, 1 black and 1 red balls and 4 white, 5 black and 3 red balls respectively. One urn is chosen at random and two balls are drawn from it. They happen to be white and red. What is the probability that they come from urns I, II and III?. (4)

4) Two dimensional random variables X and Y have joint pdf

A) $f(x,y) = \begin{cases} x+y, & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{elsewhere} \end{cases}$ find the correlation coefficient between X and Y (3)

B) If 5% of the electric bulbs manufactured by a company are defective, use Poisson distribution to find the probability that in a sample of 100 bulbs

- i. none is defective (3)
- ii. 5 bulbs will be defective .

C) The joint pdf of two continuous random variables X and Y is given as:

$$f(x,y) = \begin{cases} c(2x+y), & 2 < x < 6, 0 < y < 5 \\ 0, & \text{otherwise} \end{cases}$$

- a. What is the value of c ? (4)
- b. Write the marginal pdfs for X and Y .
- c. Find $P(X+Y > 4)$.

5) Find the moment generating function of exponential distribution and hence find its mean and variance . (3)

- A)
- B) Suppose that X is uniformly distributed over the interval $(-1, 1)$. Obtain the pdf of the random variable $Y = \sin\left(\frac{\pi X}{2}\right)$. (3)
- C) In an examination, the marks scored by the students follows the normal distribution. It is known that a student passes the examination if he secures 40% or more marks; he is placed in first, second and third division, if he secures 60% more, between 50% and 60% more, between 40% and 50% marks, respectively. He gets a distinction, if he gets 70% or more marks. It is given that 10% of the students have failed in the examination and 5% of them obtained distinction. Find the percentage of students getting second division . (4)

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Exam Date & Time: 22-Jul-2022 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

IV Semester End Semester Examination
Engineering Mathematics IV (MAT 2253)

ENGINEERING MATHEMATICS IV [MAT 2253]

Marks: 50

Duration: 180 mins.

Descriptive Questions

Answer all the questions.

Section Duration: 180 mins

- 1) A random variable X has probability mass function $P(X = k) = \frac{c}{2^k}, k = 0, 1, 2, 3, 4$
- A) i) find the value of c , ii) find $F(x)$ iii) find $P(x \text{ is even})$, and iii) find $P(x \geq 5)$. (4)
- B) There are 6 positive and 8 negative numbers. Four numbers are chosen at random, without replacement and multiplied. What is the probability that the product is a positive number. (3)
- C) In a bolt factory, machines A, B and C manufacture respectively 25%, 35% and 40% of the total. If their output 5, 4 and 2 percent are defective bolts. A bolt is drawn at random from the product and found to be defective. What is the probability that it was manufactured by machine B? (3)
- 2) The joint PDF of the random variable X , and Y is given by
- A) $f(x, y) = k(xy + y^2), 0 \leq x \leq 1, 0 \leq y \leq 2$
- Find $P(Y > 1); P\left(X > \frac{1}{2}, Y < 1\right)$ and $P(X + Y \leq 1)$. (4)
- B) Fit a least square quadratic approximation for the following data.
- | | | | | | | | |
|---|----|----|----|----|----|----|----|
| x | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| y | 80 | 90 | 92 | 83 | 94 | 99 | 92 |
- (3)
- C) A target is to be destroyed in a bombing exercise. There is 75% chance that any one bomb will strike the target. Assume that two direct hits are required to destroy the target completely. How many minimum bombs must be dropped in order that the chances of destroying the target is $\geq 99\%$? (3)
- 3) In a distribution which is exactly normal, 12% of the items are under 30 and 85% are under 60. Find the mean and standard deviation of the distribution. (4)
- A)

- B) The number of arrivals of customers during any day follows Poisson distribution with mean of 5. What is the probability that the total number of customers on two days selected at random is less than 2. (3)

- C) Apply graphical method to find the maximum value of $Z = 5x_1 + 3x_2$ subject to
 $4x_1 + 5x_2 \leq 1000; 5x_1 + 2x_2 \leq 1000; 3x_1 + 8x_2 \leq 1200; x_1 \geq 0, x_2 \geq 0$ (3)

- 4) Use simplex method to maximize $Z = 10x + 6y + 4z$, subject to

- A) $x + y + z \leq 100$
 $10x + 4y + 5z \leq 600$
 $2x + 2y + 6z \leq 300,$
 $x, y, z \geq 0.$ (5)

- B) Consider the following linear programming problem:

$$\max Z = x_1 + 3x_2 + 3x_3$$

subject to

$$x_1 + 2x_2 + 3x_3 = 4$$

$$2x_1 + 3x_2 + 5x_3 = 7,$$

(3)

find a) basic feasible solution, b) non-degenerate basic feasible solution, and c) optimal basic feasible solution.

- C) A person wants to decide the constituents of a diet which will fulfil his daily requirements of proteins, fats and carbohydrates at the minimum cost. The choice is to be made from four different types of foods. The yields per unit of these foods are given in the following table:

Food Type	Yield per unit			Cost/unit (₹)
	Proteins	Fats	Carbohydrates	
1	3	2	6	55
2	4	2	4	50
3	8	7	7	95
4	6	5	4	75
Minimum requirement	1500	1000	800	

(2)

Formulate the linear programming model for the problem.

- 5) Find the series solution of the differential equation $8x^2y'' + 10xy' - (1+x)y = 0.$ (4)

- A)
 B) Prove that $\frac{d}{dx}[x^{-n}J_n(x)] = -x^{-n}J_{n+1}(x)$, where $J_n(x)$ is the Bessel's function. (3)

- C) If $P_n(x)$ and $P_m(x)$ are Legendre's polynomials of degree n and m respectively, prove that

$$\int_{-1}^1 P_m(x)P_n(x)dx = 0, \quad n \neq m \quad (3)$$

-----End-----

Exam Date & Time: 18-Jul-2022 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

IV Semester End Semester Makeup Examination ENGINEERING MATHEMATICS - IV [MAT 2258]

Marks: 50

Duration: 180 mins.

Descriptive Questions

Answer all the questions.

- 1) Solve the boundary value problem $y'' - xy' + 3y = 11x$ with $y(1) = 1.5$, $y(2) = 15$ using the method of finite difference. Take $h = 0.25$. (4)
- A)
- B) At a telephone center, the time X (in minutes) for which an agent speaks on a telephone is found to be random variable, for which the distribution function is given by
- $$f(x) = \begin{cases} kx, & 0 \leq x \leq 2 \\ 2k, & 2 \leq x \leq 4 \\ k(6-x), & 4 \leq x \leq 6 \end{cases}$$
- a. Find the value of k , for which $f(x)$ is valid. (3)
- b. Find cumulative distribution function $F(x)$.
- c. Find $P(4 \leq X < 5 | X > 3)$.
- C) A bag A contains 2 white and 4 black balls. Another bag B contains 5 white and 7 black balls. A ball is transferred from bag A to bag B. Then a ball is drawn from the bag B. Find the probability that it is white. (3)
- 2) Solve : $\frac{\partial^2 u}{\partial t^2} = 4 \frac{\partial^2 u}{\partial x^2}$, $0 \leq x \leq 1$, $t > 0$.
- A) $u(x, 0) = \begin{cases} \frac{5x}{3}, & 0 \leq x \leq \frac{3}{5} \\ \frac{5(1-x)}{3}, & \frac{3}{5} < x \leq 1 \end{cases}$, $\frac{\partial u}{\partial t}(x, 0) = 0$ (4)
- $U(0, t) = u(1, t) = 0$, Take $h = \frac{1}{5}$. Compute $u(x, t)$ for two time steps.
- B) If $Var(X + 2Y) = 40$ and $Var(X - 2Y) = 20$,
- a. What is $Cov(X, Y)$? (3)
- b. If $Var(X) = 2Var(Y)$, what is the correlation coefficient ρ_{XY} ?

- C) A group of 9 executives include 4 who are married, 3 who never married, and 2 who are divorced. Three of the executives are to be selected for promotion. Let X denotes the number of married executives and Y denotes the number of never married executives among the 3 selected for promotion. Assuming that 3 are randomly chosen from 9 available, what will be the joint probability distribution for (X, Y) ? (3)
- 3) Solve $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 2$, $0 \leq x \leq 1.5, 0 \leq y \leq 1.5$, (4)
- A) subject to boundary conditions $u(0, y) = u(x, 0) = 0$ and $u(x, 1.5) = u(1.5, y) = 100$. Take $h=0.5$.
- B) Consider an exam that contains 10 multiple choice questions with 4 possible choices for each question, only one of which is correct. Suppose a student is to select the answer for every question randomly. Use binomial distribution to compute
- the probability for the student to get no answer correct. (3)
 - the probability for the student to get two answers correct.
 - the probability that the student fails the test, i.e., to have less than 6 correct answers.
- C) Using Z -transform, solve $y_{n+2} + 6y_{n+1} + 9y_n = 2^n$ with $y_0 = y_1 = 0$. (3)
- 4) An office has 4 secretaries handling respectively 20%, 60%, 15% and 5% of the files of a government reports. The probability that they misfile such reports are respectively 0.05, 0.1, 0.1 and 0.05. (4)
- What is the probability that a report is misfiled?
 - Find the probability that a misfiled report can be blamed on the first secretary?
- B) Find the inverse Z -transform of $\frac{2z^2+3z}{(z+2)(z-4)}$. (3)
- C) Solve: $y_{n+2} + 4y_{n+1} + 4y_n = n$. (3)
- 5) Suppose that the temperature is normally distributed with mean 50 and variance 4. What is the probability that the temperature will lie between 48°C and 53°C . (4)
- A)
- B) Suppose the random variable X has probability density function $f(x) = \begin{cases} 2x, & 0 \leq x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$. Find the probability density function of $Y = 8X^3$. (3)
- C) Suppose that the random variable X has the probability density function $f(x) = \frac{1}{2}e^{-|x|}, -\infty < x < \infty$. Obtain the moment generating function of X and hence find its mean and variance. (3)

-----End-----


FOURTH SEMESTER B.TECH. (AUTOMOBILE ENGINEERING)
MAKEUP EXAMINATIONS, JULY 2022
SUBJECT: ENGINEERING MATHEMATICS IV [MAT 2252]
Time: 3 Hours
MAX. MARKS: 50
Instructions to Candidates:

- ❖ Answer **ALL** the questions.
- ❖ Missing data may be suitably assumed.

1A.	A lot contains of 10 good articles, 4 with minor defects, 2 with major defects. Two articles are chosen at random. Find the probability that,(i) At least one is good. (ii) Neither has major defects. (iii) At most one is good.	3M												
1B.	Find the mean and variance of Exponential distribution by defining its probability distribution function.	3M												
1C.	In a normal distribution 31% of items rare under 45 and 48% are over 64. Find mean and standard deviation.	4M												
2A.	If X and Y are two random variables having joint density function $f(x,y)=\begin{cases} k(6-x-y), & 0\leq x\leq 2, \ 2\leq y\leq 4 \\ 0, & elsewhere \end{cases}$ Find k and P(X+Y < 3).	3M												
2B.	Find the mean and variance of a random variable A with the pdf $f(x) = xe^{-\frac{x^2}{2}}$, X>0.	3M												
2C.	R is a resistance to maintain a train at speed V. Find the law of the type $R = a + Vb^2$ connecting V using the following data by curve fitting method. <table><tr><td>V(miles/hour)</td><td>10</td><td>20</td><td>30</td><td>40</td><td>50</td></tr><tr><td>R(lb/ton)</td><td>8</td><td>10</td><td>15</td><td>21</td><td>30</td></tr></table>	V(miles/hour)	10	20	30	40	50	R(lb/ton)	8	10	15	21	30	4M
V(miles/hour)	10	20	30	40	50									
R(lb/ton)	8	10	15	21	30									
3A.	A random variable X is uniformly distributed over the interval -1<x<1. Find i) $P\left\{X < \frac{1}{2}\right\}$ ii) $P\left\{\left X - \frac{1}{2}\right > \frac{1}{4}\right\}$.	3M												
3B.	Solve using Graphical method Minimize $z = 3x_1 + 2x_2$ subject to $5x_1 + x_2 \geq 10$; $x_1 + x_2 \geq 6$; $x_1 + 4x_2 \geq 12$; $x_1, x_2 \geq 0$	3M												
3C.	Find the mean of the following data:	4M												



		Mid values	15	20	25	30	35	40	45	50	55	
		frequency	2	22	19	14	3	4	6	1	1	
4A.	2% of the population have a certain blood disease in serious form, 10% have in a mild form and 88% does not have it at all. A new blood test is developed. The probability that the test is positive is 0.9 if the disease in the serious form, 0.6 if in mild form, and 0.1 if no disease. A blood has tested positive. What is the probability that the blood has the serious form of the disease?											3M
4B.	Two independent random variable X and Y have mean 6 and 9 and variance 16 and 25 respectively. Find the correlation coefficient between $U = 4X - 3Y$ and $V = X + 3Y$.											3M
4C.	If X has pdf $f(x) = \lambda e^{-\lambda(x-a)}$ if $x \geq a$. Find its mgf and also find the mean and variance.											4M
5A.	A random variable X having Cauchy distribution. Show that $1/X$ also has Cauchy distribution.											3M
5B.	Suppose that $X_j, j=1, 2, \dots, 50$ are independent random variables each having a poison distribution with $\alpha = 0.03$. Let $S = X_1 + X_2 + \dots + X_{50}$ using central limit theorem, evaluate $\Pr\{S \geq 3\}$.											3M
5C.	Use simplex method to solve $\text{Maximize } Z = 10x + y$ $\text{Subject to } x + y \leq 10$ $4x + y \leq 20$ $x + 2y \leq 30$ $x, y \geq 0$											4M

Exam Date & Time: 16-Jun-2022 (09:00 AM - 12:00 PM)



MANIPAL INSTITUTE OF TECHNOLOGY
 MANIPAL
 (A constituent unit of MAHE, Manipal)

ENGINEERING MATHEMATICS IV [MAT 2257]

Marks: 50

Duration: 180 mins.

Descriptive Questions

Answer all the questions.

Section Duration: 180 mins

- 1) Use the Bender-Schmidt method to solve the following equation: $\frac{\partial u}{\partial t} = \frac{1}{2} \frac{\partial^2 u}{\partial x^2}$
- A) With respect to the conditions: $u(x, 0) = 4x - x^2, u(0, t) = u(4, t) = 0$ taking $h = 1$. (4)
 Compute u for three time steps.
- B) solve $u_{xx} + u_{yy} = 0$, $0 < x < 1$, $0 < y < 1$, given $u(0, y) = u(x, 0) = 0$,
 $u(1, y) = u(x, 1) = 50$ taking $h = 1/3$. (3)
- C) Solve the boundary value problem $y'' + y + 1 = 0$, $y(0) = y(1) = 0$, using the
 method of finite differences by taking width $h = \frac{1}{4}$. (3)
- 2) Solve the difference equation $y_{n+2} - 4y_n = n^2 + n - 1$. (4)
- A)
- B) Fit a curve $y = ab^x$ for the following data by the method of
 principle of least square
- | | | | | | | | | |
|---|---|-----|-----|-----|-----|-----|-----|-----|
| x | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| y | 1 | 1.2 | 1.8 | 2.5 | 3.6 | 4.7 | 6.6 | 9.1 |
- (3)
- C) Solve the Simultaneous equations using Z-transform method $x_{n+1} - y_n = 1$,
 $y_{n+1} + x_n = 0, x_0 = 0, y_0 = -1$. (3)
- 3) Suppose three companies X, Y and Z produce televisions. X produce twice as many as (4)

Y,

- A) while Y and Z produce the same number. It is known that 2% of X, 2% of Y and 4% of Z

are defective. All the televisions produced are put into one shop and then one television is selected at random.

- i. What is the probability that the television is defective?
- ii. Suppose a television chosen is defective, what is the probability that this television is produced by the company X? .

- B) A continuous random variable X has a pdf $f(x) = kx^2 e^{-x}$, $x \geq 0$. Find the value of

(3)

k and also find variance of X.

- C) Two absent minded roommates forget their umbrellas in some way or another. A always takes his umbrella when he goes out, while B forgets to take his umbrella with probability $1/2$. Each of them forgets his umbrella at a shop with probability $1/4$. After visiting 3 shops they return home. Find the probability that

(3)

- i. They have both the umbrellas
- ii. They have only one umbrella.

- 4) The joint pdf of two continuous random variables X and Y is given as:

A)
$$f(x, y) = \begin{cases} (x^2 + \frac{xy}{3}), & 0 \leq x \leq 1, 0 \leq y \leq 2 \\ 0, & \text{otherwise} \end{cases}$$
 (4)

Find i) The marginal pdfs for X and Y. ii) $P(X + Y \geq 1)$ iii) $P\left(Y < \frac{1}{2} \mid X < \frac{1}{2}\right)$.

- B) If $\text{Var}(X + 2Y) = 40$ and $\text{Var}(X - 2Y) = 20$,

- i. What is $\text{Cov}(X, Y)$? (3)
- ii. If $\text{Var}(X) = 2\text{Var}(Y)$, what is the correlation coefficient ρ_{XY} ?

- C) If the probability that an individual will suffer a bad reaction from injection of a given serum is 0.001 , determine the probability using Poisson's distribution that out of 2000 individuals

(3)

i) Exactly 3 individuals ii) More than 2 individuals will suffer a bad reaction.

- 5) In a normal distribution, 31% of the items are under 45 and 8% are over 64. Find the mean and standard deviation of the distribution. (4)

A)

- B) Let X_1, X_2 and X_3 are independent random variables with $M_{X_1}(t) = e^{2t(1+t)}$,
 $M_{X_2}(t) = e^{3t(1+t)}$, $M_{X_3}(t) = e^{4t(1+t)}$. Then find the pdf of $Z = 4X_1 + X_2 + 2X_3$. (3)

Hence find $E(Z/2)$ and $V(Z/2)$.

- C) If the continuous random variable X is uniformly distributed in $(-2, 2)$, find the pdf of $Y = 6 - X^2$. (3)

-----End-----

Question Paper

Exam Date & Time: 14-Jun-2022 (02:00 PM - 05:00 PM)



MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

IV Semester End Semester Examination
Engineering Mathematics IV (MAT 2253)

ENGINEERING MATHEMATICS IV [MAT 2253]

Marks: 50

Duration: 180 mins.

Descriptive Questions

Answer all the questions.

Section Duration: 180 mins

- 1) If the probability density function of a continuous random variable X is given by (4)

A)
$$f(x) = \begin{cases} ax, & 0 \leq x \leq 2 \\ (4-x)a, & 2 \leq x \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

i) find the value of a, ii) find the CDF of X, and iii) find $P(X > 2.5)$.

- B) The three bags contain 6 red, 4 black balls; 4 red, 6 black balls; 5 red, 5 black balls respectively. One of the bags is selected at random and a ball is drawn from it. If the ball drawn is red, find the probability that it is drawn from the first bag. (3)

- C) Three group of children contain respectively 3 girls and 1 boy; 2 girls and 2 boys; 1 girl and 3 boys. One child is selected at random from each group. Find the chances of selecting 1 girl and 2 boys. (3)

- 2) A 2D random variable has joint PDF (4)

A)
$$f(x, y) = \begin{cases} x^2 + \frac{xy}{3}, & 0 < x < 1, \quad 0 < y < 2 \\ 0, & \text{elsewhere} \end{cases}$$

Compute $E(X)$ and $V(Y)$.

- B) Using method of least squares, fit a curve of the form $y = ab^x$ for the following data. (3)

x	2	3	4	5	6
y	144	172.8	207.8	248.8	298.5

- C) An irregular six faced dice is thrown 12 times. The probability that it will give six even number is twice the probability that it will give five even numbers. If 1000 sets, each of exactly 12 trials are made, how many sets are expected not to give any even number? (3)

- 3) In an examination, the candidates are awarded the following grades depending on the marks scored by them: distinction: $\geq$ (4)

- A) 80%, first class: 60% $\leq$ marks < 80%, second class: 45% $\leq$ marks < 60%; third class: 30% $\leq$ marks < 45%; fail: < 30%. It

was found that 8% of the students failed and 8% have scored distinction. Find the average marks obtained by the candidates.

Deduce the percentage of students placed in second class. Assume normal distribution of marks.

- B) In a book of 600 pages, there are 60 typographical errors. Assuming Poisson law for the number of errors per page, find the probability that a randomly chosen 4 pages will contain no errors. (3)

- C) Apply graphical method to find the maximum value of $Z = 5x_1 + 3x_2$ subject to (3)

$$4x_1 + 5x_2 \leq 1000; 5x_1 + 2x_2 \leq 1000; 3x_1 + 8x_2 \leq 1200; x_1 \geq 0, x_2 \geq 0$$

- 4) Use simplex method to minimize $Z = x_1 - 3x_2 + 3x_3$, subject to (5)

A)

$$3x_1 - x_2 + 2x_3 \leq 7$$

$$2x_1 + 4x_2 \geq -12$$

$$-4x_1 + 3x_2 + 8x_3 \leq 10,$$

$$x_1, x_2, x_3 \geq 0.$$

B)

Consider the following linear programming problem:

(3)

$$\max Z = 12x_1 + 8x_2 + 14x_3 + 10x_4$$

subject to

$$5x_1 + 4x_2 + 2x_3 + x_4 = 100$$

$$2x_1 + 3x_2 + 8x_3 + x_4 = 75,$$

find a) basic feasible solution, b) non-degenerate basic feasible solution, and c) optimal basic feasible solution.

C)

A person wants to decide the constituents of a diet which will fulfil his daily requirements of proteins, fats and carbohydrates at the minimum cost. The choice is to be made from four different types of foods. The yields per unit of these foods are given in the following table:

(2)

Food Type	Yield per unit			Cost/unit (₹)
	Proteins	Fats	Carbohydrates	
1	3	2	6	45
2	4	2	4	40
3	8	7	7	85
4	6	5	4	65
Minimum requirement	800	200	700	

Formulate the linear programming model for the problem.

5)

Find the series solution near $x = 0$ of the differential equation

(4)

$$9x(1-x) \frac{d^2y}{dx^2} - 12 \frac{dy}{dx} + 4y = 0.$$

A)

B)

Prove that $\frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x)$, where $J_n(x)$ is the Bessel's function.

(3)

C)

If $P_n(x)$ is Legendre's polynomial of degree n, prove that

(3)

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n.$$

-----End-----

Exam Date & Time: 09-Jun-2022 (09:00 AM - 12:00 PM)



MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
 (A constituent unit of MAHE, Manipal)

ENGINEERING MATHEMATICS - IV [MAT 2258]

Marks: 50

Duration: 180 mins.

Descriptive Questions

Answer all the questions.

Section Duration: 180 mins

- 1) Solve the boundary value problem $y'' + xy' + y = 3x^2 + 2$ with $y(0) = 0$, $y(1) = 1$ using the method of finite difference. Take $h = 0.25$. (4)
 - A)
 - B) From a city population, the probability of selecting (i) a male or a smoker is $\frac{7}{10}$ (ii) a male smoker is $\frac{2}{5}$, and (iii) a male, if smoker is already selected is $\frac{2}{3}$. Find the probability (3) of selecting (a) a non-smoker, (b) a male, and (c) a smoker, if male is first selected.
 - C) Suppose that the life length of an electric device, X is considered as a continuous random variable with the following pdf $f(x) = ke^{-x}, x > 0$. The cost of manufacturing one such item is \$2. the manufacturer sells the item for \$5 but guarantees a total refund if $X \leq 0.9$ (3). What is the manufacturer's expected profit per item?
- 2) Solve $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = -10(x^2 + y^2 + 10), 0 < x < 3, 0 < y < 3$ with $u(x, y) = 0$ on the boundary and $h = \frac{1}{4}$. (4)
 - A)
 - B) The joint pdf of two continuous random variables X and Y is given as:

$$f(x, y) = \begin{cases} \frac{2x + y}{210}, & 2 < x < 6, 0 < y < 5 \\ 0, & \text{otherwise} \end{cases}$$
 (3)
 - a. Write the marginal pdfs for X and Y .
 - b. Find $P(X + Y > 4)$.
 - C) Suppose that X, Y & Z are three random variables with same variance 4. Also, the correlation coefficient between X & Y is $\frac{1}{4}$ and between Y & Z and Z & X is zero. What will be the correlation coefficient between U & V where $U = X + Y$, $V = Y + Z$? (3)
- 3) Compute $u(x, t)$ for four-time levels taking $h = 0.25$ and $\lambda = \frac{1}{2}$. Given $\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2}$, $0 < x < 1, t > 0$ subject to $u(x, 0) = 100(x - x^2)$, and $u(0, t) = u(1, t) = 0$. (4)

- A)
- B) If the probability that an individual will suffer a bad reaction from injection of a given serum is **0.001**, determine the probability using Poisson's distribution that out of **2000** individuals
- Exactly 3 individuals (3)
 - More than 2 individuals
- will suffer a bad reaction.
- C) Solve $y_{n+2} - 4y_n = n^2 + n - 1$. (3)
- 4) Suppose that an item is produced in three factories **X**, **Y** and **Z**. It is known that factory **X** produces thrice as many items as factory **Y**, and that factory **Y** and **Z** produce the same number of items. If 3 percent of the items produced by each of the factories **X** and **Z** are defective, while 5 percent of those manufactured by **Y** are defective. All the items produced in the three factories are stocked, and an item is selected at random. (4)
- What is the probability that the selected item is defective?
 - If an item selected is found to be defective, what is the probability that it was produced by factory **Y**?
- B) Find the **Z** - transform of $u_n = \frac{1}{n!}$ and hence evaluate $Z\left(\frac{1}{(n+1)!}\right)$. (3)
- C) Using **Z** - transform, solve $u_{n+2} + 4u_{n+1} + 3u_n = 3^n$, with $u_0 = 0, u_1 = 1$. (3)
- 5) In a normal distribution, **31%** of the items are under **45** and **8%** are over **64**. Find the mean and variance of distribution. (4)
- A)
- B) Suppose **X** is uniformly distributed over **(0,1)**, find pdf of $Y = \frac{1}{x+1}$. (3)
- C) Suppose the moment generating functions of three independent random variables X_1, X_2 and X_3 are $e^{2t(1+t)}$, $e^{3t(1+t)}$ and $e^{4t(1+t)}$ respectively. Then find the probability density function of (3)
- $$Z = 4X_1 + X_2 + 2X_3. \text{ Hence obtain } E\left(\frac{Z}{2}\right) \text{ and } V\left(\frac{Z}{2}\right).$$

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MANIPAL INSTITUTE OF TECHNOLOGY

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IV SEMESTER B.TECH (CS/IT/CC) END SEMESTER EXAMINATIONS, JUNE 2018

SUBJECT: ENGINEERING MATHEMATICS - IV

[MAT -2206]

REVISED CREDIT SYSTEM

Time: 3 Hours

MAX. MARKS: 50

Instructions to Candidates:

- ❖ Answer **ALL** questions.
- ❖ All questions carry equal marks.

1A.	Find the probability that among 7 persons: i) no two are born on the same day of the week. ii) at least two were born on the same day of the week. iii) two were born on a Sunday and two on Tuesday.	4
1B.	The chances of A, B, C becoming the general manager of a certain company are in the ratio 4:2:3. The probabilities that the bonus scheme will be introduced in the company if A, B, C become general manager are 0.3, 0.7, 0.8 respectively. If the bonus scheme has been introduced, what is the probability that A has been appointed as general manager?	3
1C.	If the random variable 'K' is uniformly distributed over [0, 5], what is the probability that the roots of the equation $4x^2 + 4xK + K + 2 = 0$ are real?	3
2A.	Compute an approximate probability that mean of a random sample of size 15 from a distribution having pdf $f(x) = \begin{cases} 3x^2, & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$ is between $\frac{3}{5}$ & $\frac{4}{5}$.	4
2B.	Two independent random variables X_1 and X_2 have means 5, 10 and variance 4, 9. Find the correlation coefficient between $U = 3X_1 + 4X_2$ and $V = 3X_1 - X_2$.	3
2C.	The income of a group of 10,000 persons was found to be normally distributed with mean Rs.750 and standard deviation of Rs. 50. Find the number of persons out of 10,000 who have income exceeding Rs.668 and those who have income exceeding Rs. 832?. Also, find the lowest income among the richest 100?	3
3A.	Find the mgf of chi-square distribution. Hence find its mean and variance.	4

3B.	The Mendelian theory states that the probabilities of classification A, B, C, D are respectively $\frac{9}{16}, \frac{3}{16}, \frac{3}{16}, \frac{1}{16}$. From a sample of 160 the actual numbers observed were 86, 35, 26 and 13. Is this data consistent with the theory at 0.01 significance level ?.	3
3C.	Let $X_1, X_2, \dots, X_n$ denote the random sample of size n from a distribution having pdf $f(x, \theta) = \begin{cases} \theta^x(1 - \theta)^{1-x}; & x = 0, 1, \dots; 0 \leq \theta \leq 1 \\ 0 & \text{Otherwise} \end{cases}$. Find the maximum likelihood estimation for θ .	3
4A.	If the random variable X has $N(\mu, \sigma^2)$ distribution, then show that the random variable $Z = \frac{X - \mu}{\sigma}$ has $N(0, 1)$ and that $V = \frac{(X - \mu)^2}{\sigma^2}$ has $\chi^2(1)$.	4
4B.	Find the mean and variance of Poisson distribution with parameter α .	3
4C.	If $\bar{X}$ is the mean of a random sample of size n from a normal distribution with mean μ and variance 100. Find n so that $P\{\mu - 5 < \bar{X} < \mu + 5\} = 0.954$.	3
5A.	Let us assume that the life length of a tire in miles, say X is normally distributed with mean θ and standard deviation 5000. Past experience indicates that $\theta = 30,000$, the manufacturer claims that the tires made by a new procedure have mean $\theta > 30,000$ and it is very possible that $\theta = 35,000$. Let us check this claim by testing $H_0 : \theta < 30,000$ against $H_1 : \theta > 30,000$. We shall observe n independent values of X say $X_1, X_2, \dots, X_n$ and we shall reject H_0 if and only if $\bar{x} \geq c$. Determine n and c so that the power function $K(\theta)$ of the test has values $K(30,000) = 0.01$ and $K(35,000) = 0.98$.	4
5B.	If 8.6, 7.9, 8.3, 6.4, 8.4, 9.8, 7.2, 7.8, 7.5 are the observed values of a random sample of size 9 from a distribution that is $N(8, \sigma^2)$, construct 90% confidence interval for σ^2 .	3
5C.	If X is a random variable and $P(x) = ab^x$, where a and b are positive such that $a + b = 1$ and $X = 0, 1, 2, \dots$. Find the moment generating function of X. Hence, show that $m_2 = m_1(2m_1 + 1)$ where m_1 and m_2 being the first two moments.	3

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MANIPAL INSTITUTE OF TECHNOLOGY

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IV SEMESTER B.TECH (CS/IT/CC) END SEMESTER EXAMINATIONS, April 2018

SUBJECT: ENGINEERING MATHEMATICS - IV

[MAT -2206]

REVISED CREDIT SYSTEM

(23/04/2018)

Time: 3 Hours

MAX. MARKS: 50

Instructions to Candidates:

- ❖ Answer **ALL** questions.
- ❖ All questions carry equal marks.

1A.	Box 1 contains 5 red and 4 green balls and box 2 contains 4 red and 6 green balls. Three balls are randomly drawn from box 1 and placed in box 2. Then a ball is taken from box 2. If the ball taken from box 2 is found red, find the probability that 2 red and 1 green balls are transferred from box 1 to box 2?	4																								
1B.	Two coins C_1 and C_2 have a probability of falling heads p_1 and p_2 , respectively. You win a bet if in 3 tosses you get at least two heads in succession. You toss the coins alternately starting with either coin. If $p_1 > p_2$, what coin would you select to start the game?	3																								
1C.	The diameter of an electric cable X is assumed to be a continuous random variable with p.d.f $f(x) = 6x(1-x)$, $0 \leq x \leq 1$. a) Find the c.d.f of X . b) Determine a number 'b' such that $P(x < b) = 2P(x > b)$	3																								
2A.	The following figures show that the distribution of digits in number chosen at random from a telephone directory. Test whether the digits occur equally frequently at 0.05 significance level. <table><tr><td>Digit</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>total</td></tr><tr><td>Frequency</td><td>1026</td><td>1107</td><td>997</td><td>966</td><td>1075</td><td>933</td><td>1107</td><td>972</td><td>964</td><td>853</td><td>10000</td></tr></table>	Digit	0	1	2	3	4	5	6	7	8	9	total	Frequency	1026	1107	997	966	1075	933	1107	972	964	853	10000	4
Digit	0	1	2	3	4	5	6	7	8	9	total															
Frequency	1026	1107	997	966	1075	933	1107	972	964	853	10000															

2B.	The random variable (X, Y) has joint pdf given by $f(x, y) = \begin{cases} x + y, & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{elsewhere} \end{cases}$. Compute the correlation coefficient between X & Y.	3
2C.	The annual rainfall at a certain locality is known to be a normally distributed random variable with mean value equal to 29.2 inches and standard deviation 2.5 inches. How many inches of rain (annually) is exceeded about 5 percent of the time?	3
3A.	Let (X ₁ , X ₂) be a sample of size 2 from the distribution having the pdf $f(x; \theta) = \begin{cases} \frac{1}{\theta} e^{-\frac{x}{\theta}} & 0 < x < \infty, \theta > 0 \\ 0 & \text{Elsewhere} \end{cases}$. We reject H ₀ : $\theta = 2$ and accept H ₁ : $\theta = 1$, if the observed values (x ₁ , x ₂) are such that $\frac{f(x_1; 2)f(x_2; 2)}{f(x_1; 1)f(x_2; 1)} \leq \frac{1}{2}$. Find significance level of the test and the power of the test when H ₀ is false.	4
3B.	Show that for the random variable X having normal distribution with mean μ and variance σ^2 , $E(X - \mu)^{2n} = 1.3.5 \dots (2n - 1)\sigma^{2n}$.	3
3C.	Suppose that random variable X is uniformly distributed over (-1, 1). Find the pdf of $Y = 4 - x^2$.	3
4A.	Let (X ₁ , X ₂ , ..., X _n) denote a random sample from a distribution which is $N(\theta_1, \theta_2)$, $-\infty < \theta_1 < \infty$, $-\infty < \theta_2 < \infty$. Find maximum likelihood estimators for θ_1 & θ_2 .	4
4B.	Find the mean and variance of Gamma distribution.	3
4C.	Suppose that X has distribution $N(\mu, \sigma^2)$. A sample of size 20 yields $\bar{x} = 3.8$ and $s^2 = 4.6$. Obtain a 90 percent confidence interval for σ^2 and μ .	3
5A.	Compute an approximate probability that the mean of a random sample of size 8, from a distribution having the pdf $f(x) = \begin{cases} 2(1-x) & 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$ is between $\frac{1}{2}$ and $\frac{5}{8}$.	4
5B.	Suppose that X is uniformly distributed over (-a, a) where a > 0. Whenever possible determine 'a' such that (i) $\Pr(X > 1) = 1/3$ (ii) $\Pr(X < 1) = 1/2$ (iii) $\Pr(X < 1) = \Pr(X > 1)$	3
5C.	Find the moment generating function of binomial distribution with parameters n and p and hence find its mean and variance.	3

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MANIPAL INSTITUTE OF TECHNOLOGY

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IV SEMESTER B.TECH. (CS/ICT/CC - ENGINEERING)

END SEMESTER MAKEUP EXAMINATIONS, JUNE-2017

SUBJECT: ENGINEERING MATHEMATICS IV [MAT 2206]

REVISED CREDIT SYSTEM

Time: 3 Hours

(16/06/2017)

MAX. MARKS: 50

Instructions to Candidates:

- ❖ Answer **ALL** the questions.
- ❖ Missing data may be suitably assumed.

1A.	Each of the two persons tosses three fair coins. Find the probability that they get the same number of heads.	3 Marks
1B.	An archer with an accuracy of 75% fires three arrows at one target. The probability of the target falling is 0.6 if it is hit once, 0.7 if it is hit twice, and 0.8 if it is hit three times. Given that the target has fallen, find the probability that it was hit twice.	3 Marks
1C.	The joint pdf of the two dimensional random variable (X,Y) is given by $f(x,y) = k xye^{-(x^2+y^2)}; x > 0, y > 0.$ (i). Find the value of k (ii). Prove that X and Y are independent.	4 Marks
2A.	Let X be a continuous random variable with pdf f given by: $f(x) = \begin{cases} ax, & 0 \leq x \leq 1, \\ a, & 1 \leq x \leq 2, \\ -ax + 3a, & 2 \leq x \leq 3, \\ 0, & \text{elsewhere.} \end{cases}$ (a) Determine the constant a . (b). Determine the cdf of X .	3 Marks
2B.	If X , Y , and Z are uncorrelated random variables with standard deviations 5, 12, and 9 respectively and if $U=X+Y$ and $V=Y+Z$, evaluate the correlation coefficient between U and V .	3 Marks
2C.	In a normal distribution 31% of the item are under 45 and 8% are over 64. Find the mean and standard deviation of the distribution.	4 Marks
3A.	Find the mean and variance of the Chi-square distribution with parameters n .	3

		Marks														
3B.	Suppose that X is uniformly distributed over the interval $(-1, 1)$. Obtain the pdf of the random variable $Y = \sin\left(\frac{\pi}{2} X\right)$.	3 Marks														
3C.	If the random variable X has pdf $f(x) = \lambda e^{-\lambda(x-a)}, x \geq a$. Find the mgf of X and hence find $E(X)$ and $V(X)$.	4 Marks														
4A.	State Central limit theorem. A sample of size 5 is obtained from a random variable with distribution $N(12, 4)$. Find the probability that the sample mean exceeds 13.	3 Marks														
4B.	A random sample of size 9 from the distribution $N(\mu, \sigma^2)$ yields $s^2 = 7.63$. Determine 95% confidence interval for σ^2 .	3 Marks														
4C.	Let $(X_1, X_2, \dots, X_n)$ be a random sample of size n from a distribution having pdf $f(x; \theta) = \begin{cases} \theta^x (1 - \theta)^{1-x}, & x = 0, 1, 2, \dots, 0 \leq \theta \leq 1 \\ 0, & \text{otherwise} \end{cases}$. Find the maximum likelihood estimator for θ .	4 Marks														
5A.	A die was cast $n = 120$ independent times and the following data resulted: <table border="1"><tr><td>Spots up</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>Frequency</td><td>b</td><td>20</td><td>20</td><td>20</td><td>20</td><td>$40-b$</td></tr></table> If we use Chi-square test, for what value of b would the hypothesis that the die is unbiased to be rejected at 0.025 significance level?	Spots up	1	2	3	4	5	6	Frequency	b	20	20	20	20	$40-b$	3 Marks
Spots up	1	2	3	4	5	6										
Frequency	b	20	20	20	20	$40-b$										
5B.	Let $(X_1, X_2, \dots, X_n)$ denote a random sample from the distribution, that has pdf $f(x; \theta) = \begin{cases} \theta x^{\theta-1}, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$. Find a best critical region for testing simple hypothesis $H_0 : \theta = 1$ against alternative hypothesis $H_1 : \theta = 2$.	3 Marks														
5C.	Let X be the random variable, having the pdf $f(x; \theta) = \begin{cases} \frac{1}{\theta} e^{-x/\theta}, & 0 < x < \infty \\ 0, & \text{otherwise} \end{cases}$. Let $H_0 : \theta = 2, H_1 : \theta = 4$. Use random sample of size 2 and define critical region to be $C = \{(X_1, X_2) / 9.5 \leq X_1 + X_2 < \infty\}$. Find the significance level of the test.	4 Marks														

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MANIPAL INSTITUTE OF TECHNOLOGY

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IV SEMESTER B.TECH. (CS/ICT/CC - ENGINEERING)

END SEMESTER EXAMINATIONS, APRIL/MAY 2017

SUBJECT: ENGINEERING MATHEMATICS IV [MAT 2206]

REVISED CREDIT SYSTEM

Time: 3 Hours

(24/04/2017)

MAX. MARKS: 50

Instructions to Candidates:

- ❖ Answer **ALL** the questions.
- ❖ Missing data may be suitably assumed.

1A.	Consider a family of n children. Let A be the event that a family has children of both sexes, and let B be the event that there is at most one girl in the family. Find the only value of n for which the events A and B are independent (assuming that each child has a probability $\frac{1}{2}$ of being a girl).	3 Marks
1B.	A bag contains 3 coins out of which one coin with 2 heads and other 2 coins are normal and unbiased. A coin is chosen at random from bag and tossed 4 times in succession. If head turns out each time, what is the probability that the chosen coin is 2 headed?	3 Marks
1C.	Suppose that the joint pdf of the two dimensional random variable (X, Y) is given by $f(x, y) = \begin{cases} x^2 + \frac{xy}{3}, & 0 < x < 1, 0 < y < 2 \\ 0, & \text{elsewhere} \end{cases}$. Compute, (i) $P(X + Y \geq 1)$ (ii) $E(X)$.	4 Marks
2A.	If the random variable K is uniformly distributed over $(0, 5)$, what is the probability that the roots of the equation $4x^2 + 4xK + K + 2 = 0$ are real?	3 Marks

2B.	Two independent random variables X_1 and X_2 have means 5 and 10, and variances 4 and 9 respectively. Find the covariance between $U = 3X_1 + 4X_2$, $V = 3X_1 - X_2$.	3 Marks
2C.	The annual rainfall at a certain locality is known to be a normally distributed random variable with mean equal to 29.5 inches and standard deviation 2.5 inches. How many inches of rain (annually) exceeded about 5% of the time?	4 Marks
3A.	Find the mean and variance of the binomial distribution with parameters n and p .	3 Marks
3B.	A random variable X has Cauchy's distribution with pdf $f(x) = \frac{1}{\pi(1+x^2)}$, $-\infty < x < \infty$. Show that $Y = \frac{1}{X}$ also has Cauchy's distribution.	3 Marks
3C.	Let $(X_1, X_2, \dots, X_n)$ be a random sample of size n from $N(\theta_1, \theta_2)$, $-\infty < \theta_1 < \infty$ and $\theta_2 > 0$. Find the maximum likelihood estimators for θ_1 and θ_2 .	4 Marks
4A.	If $X \sim N(\mu, \sigma^2)$, then show that $E[(X - \mu)^{2n}] = 1.3.5 \dots (2n-1)\sigma^{2n}$.	3 Marks
4B.	A random sample of size 17 from the distribution $N(\mu, \sigma^2)$ yields $\bar{X} = 4.7$ and $s^2 = 5.76$. Determine 90% confidence interval for μ .	3 Marks
4C.	A computer in adding numbers, rounds each number off to the nearest integer. Suppose that all rounding errors are independent and uniformly distributed over $(-0.5, 0.5)$. If 1500 numbers are added, then what is the probability that the magnitude of the total error exceeds 15? How many numbers may be added together in order that the magnitude of the total error is less than 10 with probability 0.9?	4 Marks
5A.	Mendelian theory states that the probabilities of classification a, b, c, d are respectively $\frac{9}{16}, \frac{3}{16}, \frac{3}{16}, \frac{1}{16}$. From a sample of 160 the actual numbers observed were 86, 35, 26 and 13. Is this data consistent with the theory at 1% of significance level?	3 Marks

5B.	Let $(X_1, X_2, \dots, X_n)$ denote a random sample from the distribution, that has pdf $f(x; \theta) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-\theta)^2}{2}}, \quad -\infty < x < \infty.$ Find a best critical region for testing simple hypothesis $H_0 : \theta = 0$ against alternative hypothesis $H_1 : \theta = 1$.	3 Marks
5C.	Let us assume that the life of a tire in miles, say X, is normally distributed with mean θ and standard deviation 5000. Past experience indicates that $\theta = 30,000$. The manufacturer claims that the tires made by new process have mean $\theta > 30,000$ and it is very possible that $\theta = 35,000$. Let us check his claim by testing $H_0 : \theta < 30,000$ against $H_1 : \theta > 30,000$. We shall observe n independent values of X say $X_1, X_2, \dots, X_n$, and we shall reject H_0 (thus accept H_1) if and only if $\bar{X} \geq C$. Determine n and C so that the power function $k(\theta)$ of the test has the values $k(30,000) = 0.01$ and $k(35,000) = 0.98$.	4 Marks